RIGHT TRIANGLE TRIGONOMETRY

Digital SAT Style Practice Questions

DOCUMENT 1 — QUESTIONS ONLY

Easy 25% (10) | Medium 45% (18) | Hard 30% (12) | Total Questions: 40

Q1. • EASY

In right triangle ABC, angle C is the right angle, BC = 6, AC = 8, and AB = 10. What is sin(A)?

A) 3/5
B) 4/5
C) 3/4
D) 4/3

Q2. • EASY

In right triangle DEF, angle D is the right angle, EF = 13, DE = 5, and DF = 12. What is cos(F)?

A) 5/13
B) 12/13
C) 5/12
D) 12/5

Q3. • EASY

In right triangle GHI, angle G is the right angle, GH = 9, GI = 12, and HI = 15. What is tan(H)?

A) 4/5
B) 3/5
C) 4/3
D) 3/4

Q4. • EASY

In a 45-45-90 triangle, each leg has length 9. What is the length of the hypotenuse?

A) 9
B) 18
C) $9/\sqrt{2}$
D) $9\sqrt{2}$

Q5. • EASY ✎ GRID-IN

In a right triangle, one leg has length 20 and the hypotenuse has length 29. What is the length of the other leg?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q6. ● EASY

In a 30-60-90 triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

- **A)** 14
- **B)** $7\sqrt{3}$
- **C)** 7/2
- **D)** 21

Q7. ● EASY

In a right triangle, $\sin(40°) = \cos(x°)$, where $0 < x < 90$. What is the value of x?

- **A)** 40
- **B)** 50
- **C)** 60
- **D)** 90

Q8. ● EASY

In a right triangle, the hypotenuse has length 17 and one leg has length 8. What is the length of the other leg?

- **A)** 9
- **B)** 25
- **C)** 15
- **D)** $\sqrt{353}$

Q9. ● EASY ✎ GRID-IN

In a right triangle, one leg has length 20, the other leg has length 21, and the hypotenuse has length 29. What is the sine of the angle opposite the side of length 20? (Express your answer as a fraction or decimal.)

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q10. ● EASY

In right triangle DEF, angle F is the right angle, DE = 13, EF = 5, and DF = 12. What is $\tan(D)$?

- **A)** 12/5

B) 12/13
C) 5/13
D) 5/12

Q11. ● MEDIUM

In right triangle PQR, the right angle is at R, so angle P and angle Q are complementary. If sin(P) = 5/13, what is the value of cos(Q)?

A) 5/13
B) 12/13
C) 5/12
D) 13/5

Q12. ● MEDIUM

In right triangle XYZ, the right angle is at Z, so angle X and angle Y are complementary. The value of cos(X) is $3\sqrt{5}/8$. What is the value of sin(Y)?

A) 3/8
B) $3\sqrt{5}/8$
C) $8/(3\sqrt{5})$
D) $5\sqrt{3}/8$

Q13. ● MEDIUM ✎ GRID-IN

In a right triangle, the measure of one acute angle is θ, where sin(θ) = 8/17. The side opposite θ has length 24. What is the length of the hypotenuse?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q14. ● MEDIUM

A surveyor stands 60 feet from the base of a tower and measures the angle of elevation to the top of the tower as 50°. To the nearest foot, what is the height of the tower?

A) 46
B) 39
C) 72
D) 94

Q15. ● MEDIUM

From the top of a lighthouse that is 120 feet tall, the angle of depression to a small boat is 22°. To the nearest foot, how far is the boat from the base of the lighthouse?

- **A)** 45
- **B)** 129
- **C)** 111
- **D)** 297

Q16. ● MEDIUM

A 26-foot ladder leans against a vertical wall. The base of the ladder is 10 feet from the wall. How many feet up the wall does the ladder reach?

- **A)** 24
- **B)** 16
- **C)** 36
- **D)** ≈ 27.9

Q17. ● MEDIUM ✎ GRID-IN

In a right triangle, the two legs have lengths 9 and 40. What is the area of the triangle?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q18. ● MEDIUM

If θ is an acute angle and sin(θ) = 7/25, what is the value of cos(θ)?

- **A)** 7/25
- **B)** 24/25
- **C)** 18/25
- **D)** 7/24

Q19. ● MEDIUM

Triangle ABC is similar to triangle DEF, with angle A corresponding to angle D. In triangle ABC, sin(A) = 5/13. What is the value of sin(D)?

- **A)** 13/5
- **B)** 5/12
- **C)** 5/13
- **D)** 12/13

Q20. ● MEDIUM

In a right triangle, sin(θ) = 8/17 and cos(θ) = 15/17. What is the value of tan(θ)?

- **A)** 15/8
- **B)** 17/8

C) 17/15
D) 8/15

Q21. • MEDIUM

A flagpole casts an 18-foot shadow. At this time, the angle of elevation from the tip of the shadow to the top of the flagpole is 50°. To the nearest foot, what is the height of the flagpole?

A) 21
B) 15
C) 14
D) 12

Q22. • MEDIUM ✎ GRID-IN

In a 30-60-90 triangle, the hypotenuse has length 22. What is the length of the side opposite the 30° angle?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q23. • MEDIUM

If θ is an acute angle and sin(θ) = 9/41, what is the value of cos(θ)?

A) 9/41
B) 40/41
C) 32/41
D) 9/40

Q24. • MEDIUM

In a right triangle, the measures of the two acute angles are (3x − 6)° and (2x + 16)°. What is the value of x?

A) 18
B) 20
C) 16
D) 22

Q25. • MEDIUM

The side lengths of a 30-60-90 triangle are always in the ratio 1 : √3 : 2 (short leg : long leg : hypotenuse). A particular 30-60-90 triangle has a perimeter of 60 + 20√3. What is the length of the shortest side?

A) 15

B) 10
C) 12
D) 20

Q26. ● MEDIUM ✎ GRID-IN

A person stands 60 feet from the base of a tree and measures the angle of elevation to the top of the tree as 28°. To the nearest tenth of a foot, what is the height of the tree?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q27. ● MEDIUM

Right triangle ABC has legs of length 6 and 8. Right triangle DEF is similar to triangle ABC, with every side scaled by a factor of 5. What is the length of the hypotenuse of triangle DEF?

A) 50
B) 10
C) 70
D) 40

Q28. ● MEDIUM

If θ is an acute angle and tan(θ) = 8/15, what is the value of sin(θ)?

A) 15/17
B) 8/17
C) 8/15
D) 15/8

Q29. ● HARD

In a right triangle, the measures of the two acute angles are (5x − 20)° and (x + 38)°. If sin((5x − 20)°) = 0.643, what is the value of cos((x + 38)°)?

A) 0.357
B) 1.555
C) 0.643
D) Cannot be determined without first finding x

Q30. ● HARD

In right triangle ABC, the right angle is at C, and segment CD is the altitude from C to the hypotenuse AB, where D lies on AB. If AD = 4 and DB = 9, what is the length of CD?

A) 13

B) 36
C) 6.5
D) 6

Q31. ● HARD ✎ GRID-IN

In a right triangle, one acute angle measures 40°, and the leg adjacent to this angle has length 25. To the nearest tenth, what is the perimeter of the triangle?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q32. ● HARD

If θ is an acute angle and cos(θ) = 12/37, what is the value of sin(θ)?

A) 35/37
B) 12/37
C) 25/37
D) 12/35

Q33. ● HARD

A 20-foot ladder leans against a building, with its top reaching a point 18 feet up the wall. To the nearest degree, what angle does the ladder make with the ground?

A) 26°
B) 64°
C) 42°
D) 48°

Q34. ● HARD

In a right triangle, the two legs have lengths 3x and 4x, and the hypotenuse has length 20. What is the value of x?

A) 5
B) 4.8
C) 4
D) 3.2

Q35. ● HARD ✎ GRID-IN

A guy wire is attached to the top of a 21-foot pole and anchored to the ground. The wire makes a 35° angle with the pole itself (not the ground). To the nearest tenth of a foot, what is the length of the wire?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q36. ● HARD

In a right triangle, angle A and angle B are the two acute angles, so they are complementary. If tan(A) = 5/12, what is the value of tan(B)?

- **A)** 5/12
- **B)** 5/13
- **C)** 12/13
- **D)** 12/5

Q37. ● HARD

A 16-foot pole stands vertically in the ground. A support wire runs in a straight line from the top of the pole to a point on the ground 30 feet from the base of the pole. What is the length of the support wire?

- **A)** 34
- **B)** 46
- **C)** 14
- **D)** ≈ 25.4

Q38. ● HARD

Right triangle ABC has legs 6 and 8 and a right angle at C. Right triangle DEF is similar to triangle ABC and has a hypotenuse of length 25. If angle A corresponds to angle D, what is the length of the leg in triangle DEF that corresponds to the leg of length 6 in triangle ABC?

- **A)** 20
- **B)** 15
- **C)** 12.5
- **D)** 18.75

Q39. ● HARD ✎ GRID-IN

A kite is flying at the end of a 150-foot string that makes a 62° angle with the level ground. Assuming the string is straight, to the nearest foot, how high above the ground is the kite?

GRID-IN — *enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.*

Q40. ● HARD

A right triangle has side lengths in the ratio 3 : 4 : 5, and its perimeter is 60. What is the area of the triangle?

- **A)** 125

B) 300
C) 150
D) 187.5

— END OF QUESTIONS —